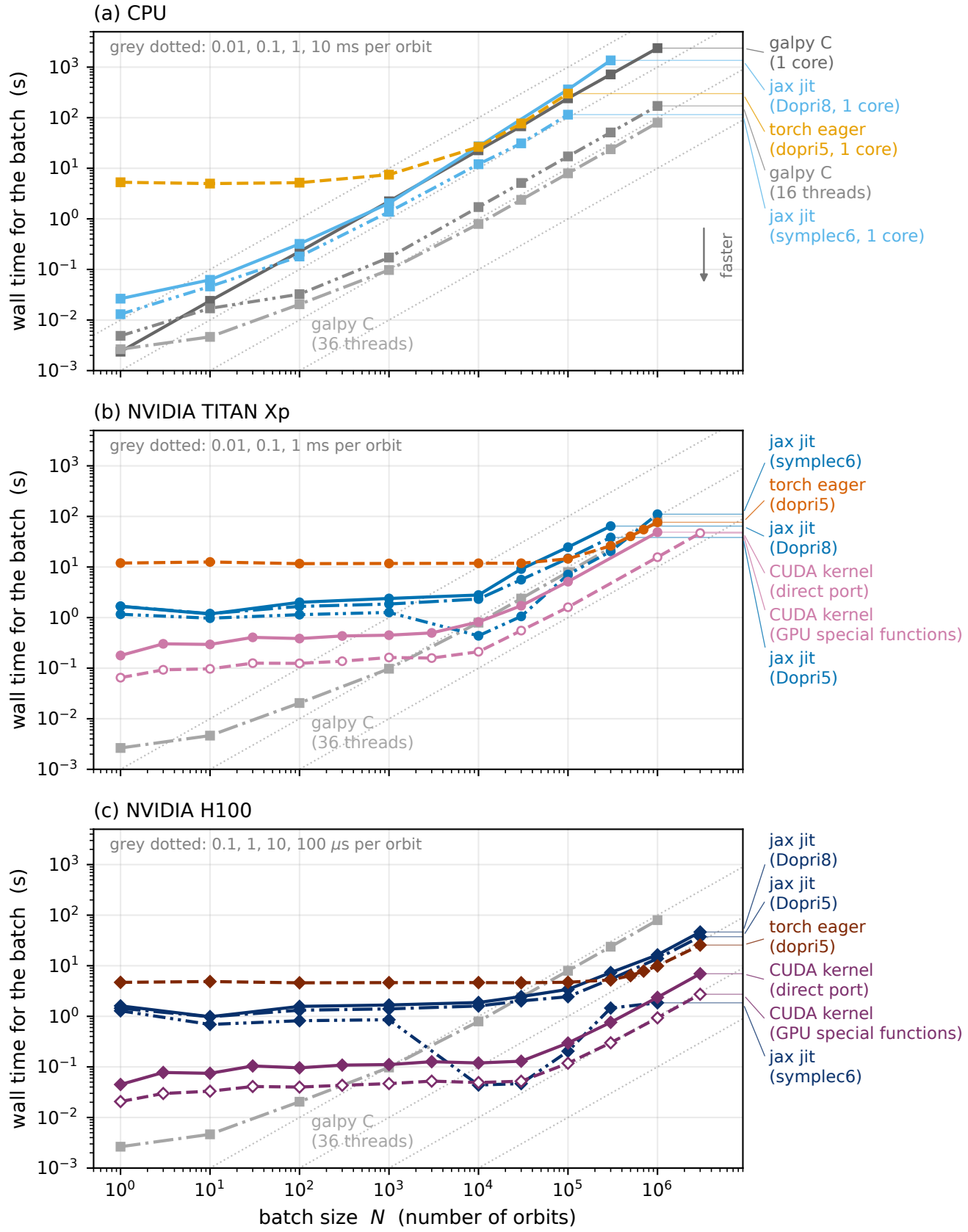
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per orbit
wall time for the batch (s)
galpy C
(1 core)
jax jit
(Dopri8, 1 core)
torch eager
(dopri5, 1 core)
galpy C
(16 threads)
jax jit
(symplec6, 1 core)
faster
galpy C
(36 threads)
(b) NVIDIA TITAN Xp
grey dotted: 0.01, 0.1, 1 ms per orbit
wall time for the batch (s)
jax jit
(symplec6)
torch eager
(dopri5)
jax jit
(Dopri8)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
jax jit
(Dopri5)
galpy C
(36 threads)
(c) NVIDIA H100
grey dotted: 0.1, 1, 10, 100 μs per orbit
wall time for the batch (s)
jax jit
(Dopri8)
jax jit
(Dopri5)
torch eager
(dopri5)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
jax jit
(symplec6)
galpy C
(36 threads)
batch size N (number of orbits)